Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

ML Engineer with 8+ years of production engineering and applied NLP/ML, specializing in **document understanding, text classification, and production model deployment** for property management and compliance domains. Built AI on FHIR — an NLP pipeline using spaCy, BERT, and semantic embeddings for natural language query translation over regulated records. Built Leasel — an agentic document understanding pipeline over 1,240 residential lease PDFs (93.6% precision, 87.9% F1). Shipped Entrata's first production ML pipeline (PyTorch + OpenAI API) across 1.6M real financial records — same property management domain, same document types, same compliance workflows as Yardi. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Shipped Entrata's **first production ML pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records — full lifecycle from research and design through output validation, quality gates, and production deployment; later productized into a customer-facing feature.
- Built **affordable housing compliance automation** (LIHTC/Tax Credit, Section 8/HCV, HUD) — income eligibility verification, AMI threshold calculations, recertification workflows, state-wise income limit imports across 100K+ unit portfolios; eliminated 80% of manual workload, zero compliance violations.
- Built and owned **20+ production modules** (Python, PostgreSQL, MySQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records — 99.5% uptime over 6+ months.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines); improved query performance 30% across 200+ SQL queries; cut execution time 50% via Docker containerization.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%.

PROJECTS

AI on FHIR: NLP Pipeline for Regulated Medical Records

Flask · spaCy · BERT · Transformers · Embeddings · LLMs · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an NLP pipeline translating natural language queries to FHIR APIs using spaCy, BERT, and semantic embeddings — **HIPAA-compliant RBAC**, ~60% reduction in retrieval time; Flask + React full-stack; demonstrates production NLP in a compliance-constrained domain directly analogous to property management records.

Leasel: Document Understanding Pipeline for Real Estate Compliance

Python · LangGraph · LLaMA · Transformers · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a production-grade document understanding pipeline over 1,240 residential lease PDFs — **93.6% precision, 82.9% recall, 87.9% F1** across 10 compliance rules, 5 US states + federal standards; deterministic rules engine keeps decisions fully auditable; pass@k eval harness (pass@1 81.7%, pass@3 88.3%).

CrashClarity: Large-Scale ML Classification Pipeline (104K Records)

Python · Scikit-learn · PyTorch · Random Forest · Logistic Regression · Feature Engineering · Pandas · R — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records — full ML lifecycle: feature engineering across weather, road, and time variables; Random Forest + Logistic Regression classifiers; Pandas data pipeline; deployed interactive analytics dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, JavaScript, TypeScript, R, Bash, HTML/CSS

ML & NLP: PyTorch, Scikit-learn, spaCy, BERT, RoBERTa, DistilBERT, Transformers, LLaMA, Embeddings, XGBoost, Random Forest, Isolation Forest, Anomaly Detection, Feature Engineering, Pandas, NumPy, Hugging Face, TensorFlow

AI & LLMs: LangGraph, LangChain, RAG Pipelines, LLM-as-Judge, LLM Evaluation, ChromaDB, FAISS, OpenAI API, Anthropic API, Ollama, Prompt Engineering, FastAPI, Flask, REST APIs

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, SQLite, Git, Linux, Agile/Scrum (120+ sprints), Tableau, Power BI